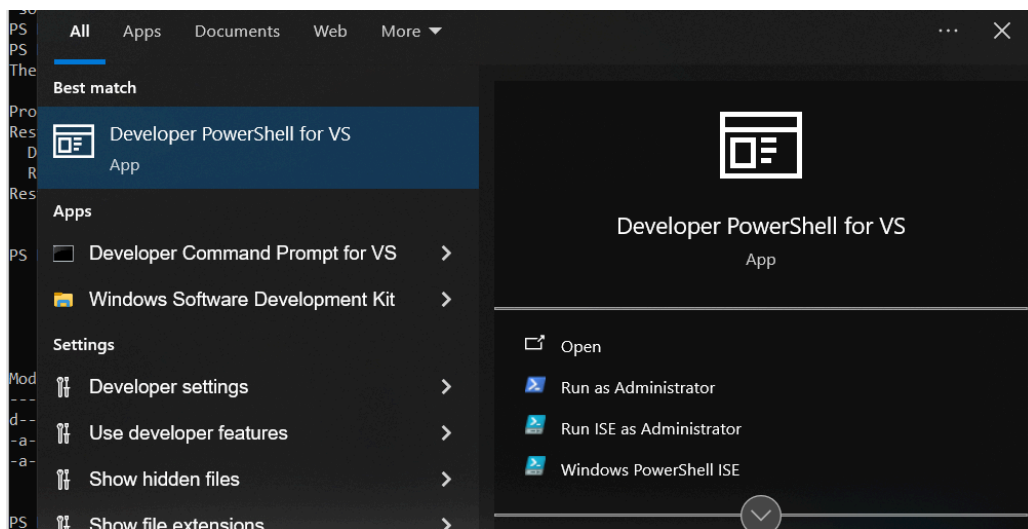


Compiling a C# script

Download and install Visual Studio

Use the Visual studio Powershell - Developer Powershell for VS - search directly



Now, navigate as you would have in Powershell ordinarily.

- If you have a current C# Project folder - you can use that by replacing the existing Program.cs with the new Program.cs which you want to compile
- Otherwise in a new directory:

```
dotnet new console --framework net8.0
```

```

solution file.
PS D:\AD-Auth-scaled\Collector\C#_collector\Debugged> ls
PS D:\AD-Auth-scaled\Collector\C#_collector\Debugged> dotnet new console --framework net8.0
The template "Console App" was created successfully.

Processing post-creation actions...
Restoring D:\AD-Auth-scaled\Collector\C#_collector\Debugged\Debugged.csproj:
  Determining projects to restore...
    Restored D:\AD-Auth-scaled\Collector\C#_collector\Debugged\Debugged.csproj (in 71 ms).
Restore succeeded.

PS D:\AD-Auth-scaled\Collector\C#_collector\Debugged> ls

Directory: D:\AD-Auth-scaled\Collector\C#_collector\Debugged

Mode                LastWriteTime         Length Name
----                -
d-----          06-Jul-26   3:21 PM             obj
-a-----          06-Jul-26   3:21 PM         252 Debugged.csproj
-a-----          06-Jul-26   3:21 PM         105 Program.cs

PS D:\AD-Auth-scaled\Collector\C#_collector\Debugged>

```

- Replace the placeholder Program.cs with the actual Program.cs

Name	Date modified	Type
obj	06-Jul-26 3:21 PM	File folder
Debugged	06-Jul-26 3:21 PM	C# Project file
Program	06-Jul-26 2:47 PM	C# Source File

```

Mode                LastWriteTime         Length Name
----                -
d-----          06-Jul-26   3:21 PM             obj
-a-----          06-Jul-26   3:21 PM         252 Debugged.csproj
-a-----          06-Jul-26   3:21 PM         105 Program.cs

PS D:\AD-Auth-scaled\Collector\C#_collector\Debugged> ls_

Directory: D:\AD-Auth-scaled\Collector\C#_collector\Debugged

Mode                LastWriteTime         Length Name
----                -
d-----          06-Jul-26   3:21 PM             obj
-a-----          06-Jul-26   3:21 PM         252 Debugged.csproj
-a-----          06-Jul-26   2:47 PM        74428 Program.cs

```

Check different size

Then command

```
dotnet build -c Release
```



In this case; compiling in the new directory did not work. So we build the script in the original directory

Errors in the new directory

```
PS D:\AD-Auth-scaled\Collector\C#_collector\Debugged> dotnet build -c Release
Determining projects to restore...
All projects are up-to-date for restore.
D:\AD-Auth-scaled\Collector\C#_collector\Debugged\Program.cs(4,14): error CS0234: The type or namespace name 'Directory
Services' does not exist in the namespace 'System' (are you missing an assembly reference?) [D:\AD-Auth-scaled\Collecto
r\C#_collector\Debugged\Debugged.csproj]
D:\AD-Auth-scaled\Collector\C#_collector\Debugged\Program.cs(401,17): error CS0246: The type or namespace name 'LdapCon
nection' could not be found (are you missing a using directive or an assembly reference?) [D:\AD-Auth-scaled\Collector\
C#_collector\Debugged\Debugged.csproj]
D:\AD-Auth-scaled\Collector\C#_collector\Debugged\Program.cs(469,13): error CS0246: The type or namespace name 'SearchS
cope' could not be found (are you missing a using directive or an assembly reference?) [D:\AD-Auth-scaled\Collector\C#_
collector\Debugged\Debugged.csproj]
D:\AD-Auth-scaled\Collector\C#_collector\Debugged\Program.cs(471,32): error CS0246: The type or namespace name 'Directo
ryControl' could not be found (are you missing a using directive or an assembly reference?) [D:\AD-Auth-scaled\Collecto
r\C#_collector\Debugged\Debugged.csproj]
D:\AD-Auth-scaled\Collector\C#_collector\Debugged\Program.cs(466,28): error CS0246: The type or namespace name 'SearchR
esultEntry' could not be found (are you missing a using directive or an assembly reference?) [D:\AD-Auth-scaled\Collect
or\C#_collector\Debugged\Debugged.csproj]
D:\AD-Auth-scaled\Collector\C#_collector\Debugged\Program.cs(508,75): error CS0246: The type or namespace name 'SearchS
cope' could not be found (are you missing a using directive or an assembly reference?) [D:\AD-Auth-scaled\Collector\C#_
collector\Debugged\Debugged.csproj]
D:\AD-Auth-scaled\Collector\C#_collector\Debugged\Program.cs(508,16): error CS0246: The type or namespace name 'SearchR
```

We replaced the original script with the debugged one and

```
dotnet build -c Release
```

```

PS D:\AD-Auth-scaled\ADAuthCollector> dotnet restore
Determining projects to restore...
Restored D:\AD-Auth-scaled\ADAuthCollector\ADAuthCollector.csproj (in 1.57 sec).
PS D:\AD-Auth-scaled\ADAuthCollector> dotnet build -c Release
Determining projects to restore...
All projects are up-to-date for restore.
D:\AD-Auth-scaled\ADAuthCollector\Program.cs(893,17): warning CA1416: This call site is reachable on all platforms. 'SecurityIdentifier.GetBinaryForm(byte[], int)' is only supported on: 'windows'. (https://learn.microsoft.com/dotnet/fundamentals/code-analysis/quality-rules/ca1416) [D:\AD-Auth-scaled\ADAuthCollector\ADAuthCollector.csproj]
D:\AD-Auth-scaled\ADAuthCollector\Program.cs(683,24): warning CA1416: This call site is reachable on all platforms. 'SecurityIdentifier.Value' is only supported on: 'windows'. (https://learn.microsoft.com/dotnet/fundamentals/code-analysis/quality-rules/ca1416) [D:\AD-Auth-scaled\ADAuthCollector\ADAuthCollector.csproj]
D:\AD-Auth-scaled\ADAuthCollector\Program.cs(1542,73): warning CA1416: This call site is reachable on all platforms. 'QualifiedAce.AceQualifier' is only supported on: 'windows'. (https://learn.microsoft.com/dotnet/fundamentals/code-analysis/quality-rules/ca1416) [D:\AD-Auth-scaled\ADAuthCollector\ADAuthCollector.csproj]
D:\AD-Auth-scaled\ADAuthCollector\Program.cs(1573,24): warning CA1416: This call site is reachable on all platforms. 'SecurityIdentifier.Value' is only supported on: 'windows'. (https://learn.microsoft.com/dotnet/fundamentals/code-analysis/quality-rules/ca1416) [D:\AD-Auth-scaled\ADAuthCollector\ADAuthCollector.csproj]
D:\AD-Auth-scaled\ADAuthCollector\Program.cs(683,24): warning CA1416: This call site is reachable on all platforms. 'SecurityIdentifier' is only supported on: 'windows'. (https://learn.microsoft.com/dotnet/fundamentals/code-analysis/quality-rules/ca1416) [D:\AD-Auth-scaled\ADAuthCollector\ADAuthCollector.csproj]
D:\AD-Auth-scaled\ADAuthCollector\Program.cs(892,38): warning CA1416: This call site is reachable on all platforms. 'SecurityIdentifier.BinaryLength' is only supported on: 'windows'. (https://learn.microsoft.com/dotnet/fundamentals/code-analysis/quality-rules/ca1416) [D:\AD-Auth-scaled\ADAuthCollector\ADAuthCollector.csproj]

```

Excellent. **The build succeeded.**

The CA1416 messages are **warnings**, not errors. They simply indicate that some APIs (`SecurityIdentifier` , `RawSecurityDescriptor` , etc.) are Windows-only. Since your collector is intended to run on Windows against Active Directory, these warnings are expected and do not prevent the build.

Your next step is to run the newly built collector from:

```
D:\AD-Auth-scaled\ADAuthCollector\bin\Release\net8.0\
```

Use the same command you used previously to collect from your large Active Directory environment.

The first thing to check in the output is whether the counts for:

- Users
- Groups
- Computers

are now above 1000. If they are, then the referral-chasing fix has likely resolved the paging issue. After that, we can verify the ACL count and continue with the rest of the pipeline if everything looks correct.

folder containing **only the executable and all required dependencies**, use **publish**, not build.

From the project directory (`D:\AD-Auth-scaled\ADAuthCollector`), run:

```
dotnet publish -c Release -r win-x64 --self-contained true
```

```
genericAce.IsInherited' is only supported on: 'windows'. (https://learn.microsoft.com/dotnet/fundamentals/code-quality-rules/ca1416) [D:\AD-Auth-scaled\ADAuthCollector\ADAuthCollector.csproj]
ADAuthCollector -> D:\AD-Auth-scaled\ADAuthCollector\bin\Release\net8.0\win-x64\ADAuthCollector.dll
ADAuthCollector -> D:\AD-Auth-scaled\ADAuthCollector\bin\Release\net8.0\win-x64\publish\
PS D:\AD-Auth-scaled\ADAuthCollector>
```

Creates all the executable, and all the dependencies in one folder